WiFi-CRO

Complete Hardware Wiring Guide

ESP8266 + ADS1115 + ACS712 + Power Supply

NIT-H Electrathon 2026 • Differential Mode • ±30V / ±30A

Voltage Range: ±30V safe (hardware limit ±51.8V via Zener protection)

Current Range: ±30A continuous (ACS712 30A model)

Frequency: DC to 200 Hz accurate

Sample Rate: 430 SPS per channel (860 SPS ADS1115 interleaved)

Resolution: Voltage ~2 mV | Current ~1.9 mA (16-bit ADS1115)

Waveform: Full signed ± — both positive and negative halves captured

Interface: Phone hotspot → ESP8266 connects → Laptop opens index.html

Total Cost: ≈ ₹774 complete

1. Component List

COMPONENT	SPEC	QTY	PRICE	PURPOSE
NodeMCU ESP8266	v3, CH340G	1	₹200	Main controller + WiFi
ADS1115 module	16-bit, I2C, 860SPS	1	₹120	Precision ADC for both channels
ACS712 module	30A version	1	₹80	Current sensor — measures amps
Resistor 100kΩ	¼W, 1%	1	₹2	Voltage divider upper R1 part A
Resistor 47kΩ	¼W, 1%	1	₹2	Voltage divider upper R1 part B
Resistor 10kΩ	¼W, 1%	1	₹2	Voltage divider lower R2
Zener Diode 3.3V	1N4728A or BZX55C3V3	1	₹5	Clamps ADS1115 input — protection
Capacitor 0.1µF	Ceramic, 50V	2	₹4	Filter caps — noise suppression
LiPo Battery	3.7V, 1000–2000mAh	1	₹120	Portable power source
TP4056 module	Type-C charging	1	₹14	LiPo charger
MT3608 boost	5V boost converter	1	₹25	3.7V → 5V for NodeMCU

Breadboard	400 tie, half size	1	₹60	Prototyping base
Jumper wires	M-M and M-F	20+	₹40	Connections
Probe wires	With alligator clips	4	₹30	RED, BLACK, YELLOW, ORANGE probes
			≈ ₹774	TOTAL ESTIMATED COST

ACS712 Note: Use 30A model for this guide. If using 5A model: change ACS_SENSITIVITY = 0.185 in firmware. 20A model: use 0.100.

2. Circuit Overview

The circuit has four subsystems that connect together:

① POWER	② VOLTAGE	③ CURRENT	④ BRAIN
LiPo → TP4056 → MT3608 → 5V	RED/BLACK probes → Divider → ADS1115	YELLOW/ORANGE → ACS712 → ADS1115	ADS1115 → I2C → ESP8266 → WiFi

3. Voltage Divider Circuit (Subsystem ②)

The voltage divider scales down the input signal from ±30V to within the ADS1115 safe range of ±4.096V.

3.1 Component Values

COMPONENT	VALUE	FORMULA	RESULT
R1 (upper)	100kΩ + 47kΩ = 147kΩ	Two resistors in series	147,000 Ω
R2 (lower)	10kΩ	Single resistor	10,000 Ω
V_SCALE	(R1+R2) / R2	(147k+10k) / 10k	**= 15.7 ✓**

Max input voltage	Zener clamps at 3.3V	3.3V × 15.7	**= 51.8V safe**

3.2 Voltage Divider Wiring Diagram

```
                    VOLTAGE DIVIDER CIRCUIT


RED probe ──────────────────────────────────────┐
(circuit +)         R1a = 100kΩ                 │
                ┌──///──┐                     │
                │           │  R1b = 47kΩ       │
                │       ┌──///──┐           │
                │       │           │           │
┌───────────────┘       └───────────┤           │
│                                   │           │
│              ┌────── NODE A ──────┤           │
│              │     (junction of   │           │
│              │      R1 and R2)    │           │
│              │                    │           │
│         ADS1115 A0(+)        R2 = 10kΩ        │
│              │             ┌──///──┐         │
│              │             │         │         │
│              │             │         │         │
│         ADS1115 A1(-)──────┤         │         │
│              │             │       BLACK probe──┘
│              │             │       (circuit -)
│          GND ──────────────┘
│
└── 3.3V Zener (cathode to NODE A, anode to BLACK probe)
└── 0.1µF cap  (NODE A to BLACK probe  — noise filter)


DIFFERENTIAL: ADS1115 measures  A0(+) minus A1(-)
Result is fully signed  —  positive AND negative volts captured
```

4. ACS712 Current Sensor (Subsystem ③)

The ACS712 senses current by measuring the magnetic field around a conductor. No direct electrical connection to the load — safe and isolated.

4.1 ACS712 Pin Map

PIN	NAME	CONNECTS TO
1	VCC (5V)	MT3608 5V output (the boost converter output)
2	GND	Common GND (same ground as ESP8266 and ADS1115)
3	VIOUT	→ 0.1μF cap to GND → ADS1115 A2 (current signal output)
4	IP+	YELLOW probe (current enters here from circuit under test)
5	IP−	ORANGE probe (current exits here back to circuit under test)

4.2 Current Sensor Wiring Diagram

```
                    ACS712 CURRENT SENSOR


YELLOW probe ──────── ACS712 IP+ ─── current path ─── ACS712 IP- ──────── ORANGE probe

(into circuit)                  (through sensor)                    (out of circuit)


ACS712 module:

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│  VCC  ──────────────── MT3608 5V output  │

│  GND  ──────────────── Common GND        │

│  VIOUT ─────────────── 0.1µF ─── GND     │  (noise filter cap)

│              │                           │

│              └──────── ADS1115 A2(+)      │  (current signal in)

└──────────────────────────────────────────┘


ADS1115 A3 ──────────────────── GND            (current reference)


DIFFERENTIAL: ADS1115 measures  A2(+) minus A3(-)


ACS712 output at rest (0 Amps):   VIOUT = 2.5V

ACS712 output at +10A:            VIOUT = 3.16V  (2.5 + 10×0.066)

ACS712 output at -10A:            VIOUT = 1.84V  (2.5 - 10×0.066)

Firmware formula:  Amps = (VIOUT - 2.5) / 0.066
```

5. ADS1115 ADC Module (Subsystem ④)

The ADS1115 is the heart of the measurement system. It is a 16-bit differential ADC with I2C interface, running at 860 samples per second.

5.1 ADS1115 Pin Map

PIN	NAME	CONNECTS TO
VDD	Power	NodeMCU 3.3V pin

GND	Ground	NodeMCU GND (common ground)
SCL	I2C Clock	NodeMCU D1 (GPIO5)
SDA	I2C Data	NodeMCU D2 (GPIO4)
ADDR	I2C Address	GND → sets address to 0x48 (default)
ALRT	Alert	Not connected (leave floating)
A0	Diff+ Voltage	Node A of voltage divider (after R1, before R2)
A1	Diff− Voltage	BLACK probe (circuit ground / negative reference)
A2	Diff+ Current	ACS712 VIOUT (through 0.1µF filter cap)
A3	Diff− Current	GND (current measurement reference)

Why Differential? Single-ended mode (A0 vs GND) can only show positive voltages. Differential mode (A0 minus A1) gives a fully signed result — captures the complete AC waveform including negative half. No capacitor needed, no signal distortion.

5.2 ADS1115 Configuration (in firmware)

SETTING	VALUE	REASON
Gain	GAIN_ONE	±4.096V input range — fits R2 voltage (max 3.3V)
Data Rate	860 SPS	Maximum speed — interleaved gives 430 SPS per channel
Voltage read	Differential 0_1	A0(+) minus A1(−) — fully signed voltage
Current read	Differential 2_3	A2(+) minus A3(−) — fully signed current
I2C Address	0x48	ADDR pin tied to GND (default)

6. NodeMCU ESP8266 Pin Map

NodeMCU PIN	GPIO	LABEL	CONNECTS TO
3V3	—	3.3V output	ADS1115 VDD (power for ADC module)
GND	—	Ground	ADS1115 GND + ADDR | ACS712 GND | R2 bottom | A1 | A3
VIN / VU	—	5V input	MT3608 5V output (powers the whole board)
D1	GPIO5	SCL	ADS1115 SCL (I2C clock)

D2	GPIO4	SDA	ADS1115 SDA (I2C data)
D3–D8	Various	Unused	Not connected (keep free for future expansion)

7. Power Supply Chain (Subsystem ①)

Three-stage portable power: LiPo battery → TP4056 charger module → MT3608 boost converter → 5V rail → all components.

```
                    POWER SUPPLY CHAIN


LiPo 3.7V ——— TP4056 B+/B- ——— TP4056 OUT+/OUT-

battery         (charging module)        |

                Type-C USB to charge     |

                                         |

                           MT3608 IN+/IN-

                           (boost converter)

                           Adjust pot until OUT = 5.0V

                                         |

                           MT3608 OUT+ ─┬─────────────────┐

                                         |                 |

                                 NodeMCU VIN        ACS712 VCC

                                         |

                           MT3608 OUT- ─┴─────────────────┐

                                   (GND)           ACS712 GND


NodeMCU 3.3V pin ———————————————————— ADS1115 VDD

(internal 3.3V regulator on NodeMCU)      (3.3V only — do NOT

                                           connect to 5V!)


IMPORTANT: Use multimeter to set MT3608 output to exactly 5.0V

before connecting to NodeMCU. Too high will damage the board.
```

8. Complete Wiring Table

Every single wire in the circuit listed in order. Follow this table exactly.

8.1 Power Connections

WIRE	FROM	TO	NOTE

RED	LiPo B+	TP4056 B+	Battery positive to charger
BLACK	LiPo B−	TP4056 B−	Battery negative to charger
RED	TP4056 OUT+	MT3608 IN+	Charger output to boost input
BLACK	TP4056 OUT−	MT3608 IN−	Charger GND to boost GND
RED	MT3608 OUT+	NodeMCU VIN	5V boost to NodeMCU power
RED	MT3608 OUT+	ACS712 VCC	5V to current sensor
BLACK	MT3608 OUT−	NodeMCU GND	Common GND rail
BLACK	MT3608 OUT−	ACS712 GND	Common GND to current sensor
ORANGE	NodeMCU 3V3	ADS1115 VDD	3.3V to ADC — NOT 5V!
BLACK	NodeMCU GND	ADS1115 GND	GND to ADC
BLACK	NodeMCU GND	ADS1115 ADDR	Sets I2C address to 0x48

8.2 I2C Bus Connections (Brain)

WIRE	FROM	TO	NOTE
YELLOW	NodeMCU D1 (GPIO5)	ADS1115 SCL	I2C clock line
WHITE	NodeMCU D2 (GPIO4)	ADS1115 SDA	I2C data line

8.3 Voltage Measurement Connections

WIRE	FROM	TO	NOTE
RED	RED probe tip	R1a 100kΩ end 1	Probe to first resistor
RED	R1a 100kΩ end 2	R1b 47kΩ end 1	R1a series to R1b
RED	R1b 47kΩ end 2	ADS1115 A0	Junction: top of R2, input to ADC
ORANGE	R1b 47kΩ end 2	R2 10kΩ end 1	Same junction: top of R2
BLACK	R2 10kΩ end 2	ADS1115 A1	Bottom of R2 → ADC A1
BLACK	R2 10kΩ end 2	BLACK probe tip	Same point: circuit reference
ORANGE	ADS1115 A0	Zener CATHODE	Protection: cathode to signal
BLACK	BLACK probe / GND	Zener ANODE	Protection: anode to GND
ORANGE	ADS1115 A0	0.1μF cap top	Noise filter: cap to signal node
BLACK	GND	0.1μF cap bottom	Noise filter: cap to GND

8.4 Current Measurement Connections

WIRE	FROM	TO	NOTE
YELLOW	YELLOW probe tip	ACS712 IP+	Current enters sensor
ORANGE	ACS712 IP−	ORANGE probe tip	Current exits sensor
GREEN	ACS712 VIOUT	0.1μF cap top	Signal to noise filter
GREEN	0.1μF cap top	ADS1115 A2	Filtered signal to ADC
BLACK	0.1μF cap bottom	GND	Filter cap to GND
BLACK	GND	ADS1115 A3	Current reference GND

9. Complete Schematic (All Connections)

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║                    WiFi-CRO COMPLETE SCHEMATIC                        ║
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┌──────────────────────────────────────────────────────────────────────┐
│  POWER CHAIN                                                         │
│  LiPo 3.7V ── TP4056 ── MT3608(adj→5.0V) ─┬── NodeMCU VIN            │
│                                           └── ACS712 VCC             │
│  NodeMCU 3V3 ──────────────────────────── ADS1115 VDD                │
│  GND (common) ─────────────────────────── ADS1115 GND+ADDR           │
│                │                          └── ACS712 GND             │
└────────────────┼─────────────────────────────────────────────────────┘


┌────────────────┼─────────────────────────────────────────────────────┐
│  I2C BUS       │                                                     │
│  NodeMCU D1 (GPIO5) ──────────────────── ADS1115 SCL                 │
│  NodeMCU D2 (GPIO4) ──────────────────── ADS1115 SDA                 │
└────────────────┼─────────────────────────────────────────────────────┘


┌────────────────┼─────────────────────────────────────────────────────┐
│  VOLTAGE MEASUREMENT                                                 │
│                                                                      │
│  RED probe ──[100kΩ]──[47kΩ]──┬──── ADS1115 A0 (+)                   │
│                               │                                      │
│                             [10kΩ]  [Zener 3.3V]  [0.1µF]            │
│                               │    (A0→GND)       (A0→GND)           │
│  BLACK probe ─────────────────┴──── ADS1115 A1 (−)                   │
│                               │                                      │
│                              GND                                     │
│                                                                      │
│  DIFFERENTIAL: reads A0 − A1  →  multiply × 15.7  =  real volts      │
└──────────────────────────────────────────────────────────────────────┘


┌──────────────────────────────────────────────────────────────────────┐
│  CURRENT MEASUREMENT                                                 │
│                                                                      │
```

10. Breadboard Layout Guide

Suggested physical placement on a 400-tie half-size breadboard.

```
BREADBOARD LAYOUT  (top view, 400-tie half-size)


┌──────────────────────────────────────────────────────────┐
│  +  + + + + + + + + + + + + + + + + + + + + + + + + + +  │  ← 5V rail
│  -  - - - - - - - - - - - - - - - - - - - - - - - - - -  │  ← GND rail
│                                                          │
│  Rows 1-15:   NodeMCU ESP8266  (straddles centre gap)    │
│               Place across columns A-J                   │
│                                                          │
│  Rows 17-20:  ADS1115 module  (4 pins wide)              │
│               VDD→3V3, GND→rail, SCL→D1, SDA→D2          │
│               A0-A3 available on right side              │
│                                                          │
│  Rows 22-25:  ACS712 module   (5 pins)                   │
│               VCC→5V rail, GND→rail                      │
│               IP+ / IP- → probe wires (exit breadboard)  │
│               VIOUT → 0.1µF → A2                         │
│                                                          │
│  Rows 27-28:  R1a 100kΩ  (col E to col J)                │
│  Row  29:     R1b  47kΩ  (col E to col J)                │
│  Row  30:     R2   10kΩ  (col E to GND rail)             │
│               Junction = row 29 col J = A0 input         │
│                                                          │
│  Row  31:     Zener 3.3V  (cathode→junction, anode→GND)  │
│  Row  32:     0.1µF cap voltage  (junction → GND rail)   │
│  Row  33:     0.1µF cap current  (VIOUT node → GND)      │
│                                                          │
│  MT3608 + TP4056:  mount outside breadboard, wire in     │
│  LiPo:             underneath or beside breadboard       │
│                                                          │
│  -  - - - - - - - - - - - - - - - - - - - - - - - - - -  │  ← GND rail
│  +  + + + + + + + + + + + + + + + + + + + + + + + + + +  │  ← 5V rail
└──────────────────────────────────────────────────────────┘
```

11. Step-by-Step Build Order

Follow this exact sequence. Test at each step before continuing.

STEP 1 — Set MT3608 voltage FIRST (before connecting anything)

Power MT3608 from TP4056 only. Use multimeter on output terminals. Adjust trimmer pot until output reads exactly 5.0V. This is critical — too high destroys NodeMCU.

STEP 2 — Power rail wiring

Connect MT3608 OUT+ to breadboard top + rail. Connect GND to both − rails. Connect NodeMCU VIN to 5V rail and GND. Power on — NodeMCU LED should light.

STEP 3 — I2C wiring

NodeMCU D1 (GPIO5) → ADS1115 SCL. NodeMCU D2 (GPIO4) → ADS1115 SDA. NodeMCU 3V3 → ADS1115 VDD. NodeMCU GND → ADS1115 GND and ADDR.

STEP 4 — Upload firmware and test I2C

Upload esp8266_firmware.ino. Open Serial Monitor at 115200 baud. Should print 'ADS1115 OK' and then show the ESP's IP address. If it prints 'ADS1115 not found' — check SDA/SCL wires are not swapped.

STEP 5 — Build voltage divider

Place 100kΩ and 47kΩ in series (R1). Place 10kΩ (R2) from junction to GND rail. Junction point connects to ADS1115 A0. Wire BLACK probe rail to ADS1115 A1 and to bottom of R2.

STEP 6 — Add protection components

Zener diode: cathode to A0 junction, anode to GND. 0.1μF cap: across same two points (A0 junction to GND). These protect ADS1115 from overvoltage.

STEP 7 — Test voltage channel

Connect RED probe to a 9V battery +. Connect BLACK probe to battery −. Open webpage, enter ESP IP. Sidebar should show Vdc ≈ 9V. Fine-tune V_SCALE in firmware if reading is slightly off.

STEP 8 — Wire ACS712

ACS712 VCC → 5V rail. ACS712 GND → GND rail. ACS712 VIOUT → 0.1μF cap → ADS1115 A2. GND → ADS1115 A3. YELLOW and ORANGE probe wires to IP+ and IP−.

STEP 9 — Test current channel

With YELLOW and ORANGE probes disconnected — ACS712 VIOUT = 2.5V = reads 0.00A. Webpage current channel should show flat line at 0A. Connect probes in series with a load (e.g. LED + resistor from battery) and observe current reading.

STEP 10 — Full system test

Connect a 3-0-3 transformer (use full 6V winding). Voltage tab should show sine wave ±8.5V, 50Hz. FFT should show single spike at 50Hz. Crest factor ≈ 1.414. AI should identify 'AC Transformer'.

12. Firmware Configuration

Only 2 lines need changing in the firmware:

WIFI_SSID → Your phone hotspot name (line ~20)

WIFI_PASS → Your phone hotspot password (line ~21)

Everything else → Leave exactly as-is. V_SCALE, ACS_SENSITIVITY, all pins are pre-configured.

CONSTANT	VALUE / DESCRIPTION
V_SCALE	15.7 → (100k+47k+10k) / 10k = 157k/10k
ACS_OFFSET	2.5 → ACS712 VIOUT at 0 amps
ACS_SENSITIVITY	0.066 → 30A model. Use 0.100 for 20A, 0.185 for 5A
SAMPLE_RATE	860 SPS → 430 per channel → 1 pair every 2.32ms
SCL pin	GPIO5 = D1 on NodeMCU board
SDA pin	GPIO4 = D2 on NodeMCU board
ADS1115 ADDR	0x48 → ADDR pin tied to GND

13. Demo Circuit Connections

Three circuits to demonstrate at the hackathon, from easiest to most impressive.

Demo A — Battery (DC constant)

RED probe	Battery + (9V or AA pack)
BLACK probe	Battery −
Expected	Flat DC line. Vdc ≈ battery voltage. Vpp < 0.1V. Freq = 0 Hz. AI: 'Battery (no load)'

Demo B — 555 Timer Square Wave

555 circuit	9V battery, 10kΩ + 47kΩ×2 + 10μF + 0.01μF + LED+330Ω → ~1Hz square
RED probe	Output pin 3 of 555
BLACK probe	GND of circuit
Expected	Square wave 0–9V. Crest factor ≈ 1.0. Freq ≈ 1Hz. AI: 'Square Wave Oscillator'

Demo C — AC Transformer (most impressive)

Transformer	3-0-3 step-down. Use full 6V winding (outer-to-outer terminals). Load: 220Ω resistor.
RED probe	One outer terminal of transformer secondary
BLACK probe	Other outer terminal (NOT centre tap)
YELLOW/ORANGE	In series with the 220Ω load resistor
Expected	Full sine wave ±8.5V. Freq = 50Hz. Crest ≈ 1.414. Both halves visible. AI: 'AC Transformer'

14. Troubleshooting

SYMPTOM	FIX
ADS1115 not found	Check D1=SCL and D2=SDA — they are often swapped. Check 3.3V on VDD.
Voltage reads 0V	Check R1/R2 divider wired correctly. Check A1 connected to BLACK probe. Check GAIN_ONE set.
Voltage reads wrong value	Measure actual voltage with multimeter. Adjust V_SCALE in firmware: V_SCALE = multimeter_reading / ads_reading.
AC shows only half wave	A1 must go to BLACK probe, not GND. Must use readADC_Differential_0_1 not SingleEnded.
Current always 0A	ACS712 needs 5V — check VCC connected to 5V rail not 3.3V. Check VIOUT connected to A2.
Current reads ~38A at rest	ACS712 VCC is too low or missing. Needs exactly 5V. If VIOUT = 0V instead of 2.5V, chip has no power.
Cannot reach ESP IP	Laptop and ESP must be on same WiFi. Check SSID/password in firmware. IP shown in Serial Monitor at 115200.
Graph shows 0–1ms X axis	This is fixed in latest index.html. Re-download the file. X axis updates on sample button click.
AI button greyed out	AI button enables only after first successful data capture. Wait for CAPTURE COMPLETE status.
Noisy / spiky waveform	Add 0.1μF caps if not already present. Ensure solid GND connections. Keep probe wires short.
NodeMCU not booting	Check MT3608 output is exactly 5.0V. Lower than 4.5V = won't boot. Higher than 5.5V = damaged.

15. Quick Reference Card

KEY VALUES	PROBE CONNECTIONS
V_SCALE = 15.7	RED → Circuit positive (+)
ACS_OFFSET = 2.5V	BLACK → Circuit negative / GND
ACS_SENSITIVITY = 0.066 (30A)	YELLOW → Current path IN (IP+)
Sample rate = 430 SPS/channel	ORANGE → Current path OUT (IP−)
dt per sample = 2.32 ms	
Voltage safe max = ±51.8V	For voltage only: RED + BLACK
Current rated max = ±30A	For current only: YELLOW + ORANGE
I2C address = 0x48	For both: all 4 probes
SCL = D1 (GPIO5)	
SDA = D2 (GPIO4)	NEVER connect RED probe to mains AC
	Safe max input voltage = ±51.8V

⚠ **SAFETY:** Never connect probes to mains 230V AC — maximum safe voltage is ±51.8V. Always set MT3608 to exactly 5.0V before powering NodeMCU. Double-check all GND connections are common before powering on.

✓ **DONE CHECKLIST:** MT3608 set to 5.0V ✓ | ADS1115 found in Serial Monitor ✓ | Voltage reads correctly with multimeter ✓ | Current reads 0A at rest ✓ | ESP IP shown in Serial Monitor ✓ | index.html opens and shows data ✓